

Constrained Decoding for Gender-Fair French Language Generation in LLMs

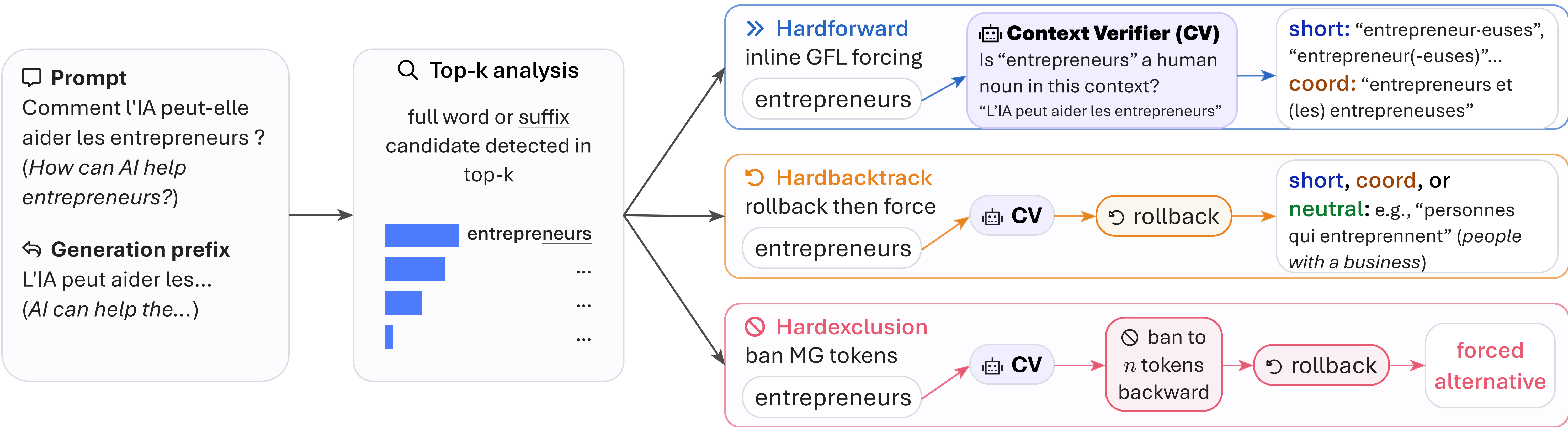
Enzo Doyen & Amalia Todirascu (LiLPa, University of Strasbourg)

1. Introduction »

- Gender bias long studied in NLP, lately focusing on biases stemming from interactions with LLMs^[1,2]
- One form of understudied gender bias in gender-marked languages (e.g. French, German, Polish) is the **masculine generics (MG)** bias, i.e. **the use of the masculine form as a default/neutral form to refer to mixed group of men/women**. Example in French:

Les étudiants^{MG} sont partis. (Students^{MG} left.)

- Psycholinguistics studies have shown that **MG induce bias by amplifying male-centric mental representations**^[3–5]
- In generic conversational contexts, **LLMs also use MG**^[6,7] and are **reluctant to using gender-fair language (GFL)**^[6]. Existing mitigation to reduce MG and promote GFL (e.g. gender rewriting^[8,9]) operates only via post-processing, which fails to prevent the initial generation of biased text
- We propose a set of inference-based systems to promote GFL in French



3. Preliminary Findings

- Tested with Llama 3.1 (8B), Ministral 3 (14B), EuroLLM (22B) & Qwen 3.5 (35B); Qwen 3.5 (2B) as context verifier model
- Example with prompt “How much do students earn?” (Combien les étudiants gagnent-ils ?)

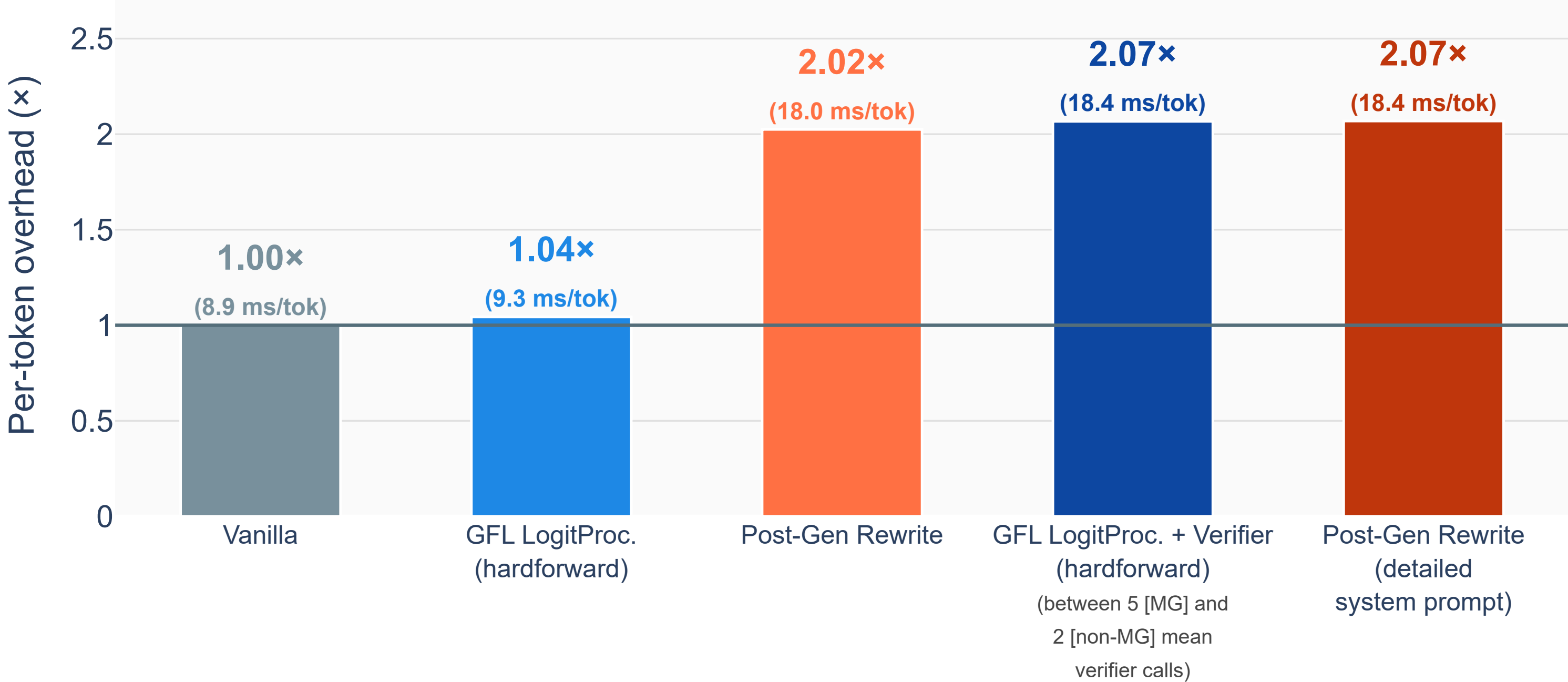
Vanilla (Llama 3.1)	Constrained (hardforward, short)
Le montant des allocations pour les étudiants en France peut varier en fonction de plusieurs facteurs, tels que la région d’habitation [...] Cependant, en France, les étudiants peuvent être éligibles à plusieurs types de bourses d’études [...]	Le montant des allocations pour les étudiant-es en France varie en fonction de plusieurs facteurs tels que la région d’habitation [...] Cependant, en France, les étudiant-es peuvent bénéficier d’une aide financière appelée bourse études [...]

- Interestingly, initial GFL forcing is sometimes enough to nudge the model to generate GFL by itself (e.g. when the constraint is applied at the beginning (first 1/2 forcings) and later fully removed)

- Performance without verifier close to vanilla model; with verifier, close to double pass on same model for rewriting

Per-token decoding overhead relative to Vanilla

Llama-3.1-8B-Instruct | NVIDIA H200 | 30 MG prompts + 30 non-MG prompts | mean of 10 runs per condition



2. Methodology

- 3 main constrained decoding systems** with 3 different strategies a/b/c:
 - a. short:** force gender-fair short suffix (e.g., ‘étudiant·es’)
 - b. coord:** force coordinated gendered pairs (e.g. ‘étudiants et étudiantes’) (→ different forcing modes to force (1) conjunction only, (2) conjunction + first token or (3) conjunction + full paired word)
 - c. neutral:** force gender-neutral form (e.g. ‘corps étudiant’ [*student body*])

System	Mechanism	Strategies
hardforward	Forces MG about to be generated into gender-fair form	short, coord
hardbacktrack	Backtracks after a MG has been generated and replaces it with a gender-fair form	short, coord, neutral
hardexclusion	Bans MG tokens with NoBadWordsLogitsProcessor	—

4. Prospects

- Continued development of all 3 systems, particularly hardbacktrack and hardexclusion
- Future evaluation of all systems through a three-dimensional Pareto analysis:

$$\hat{M} = \frac{MG}{HN} \quad \hat{I} = \frac{I}{HN} \quad \hat{N} = \frac{N}{HN}$$

where HN = human noun, *I* = inclusive form, and *N* = neutral form; minimizing $\hat{M}$ and maximizing $\hat{I}$, $\hat{N}$

- Comparison with vanilla models to evaluate coherent generation and LLM instruction-following abilities
- Comparison with models fine-tuned on gender-fair instruction responses

References

[1] Kotek, H. et al. (2023). Gender Bias and Stereotypes in Large Language Models. In *Proceedings of The ACM Collective Intelligence Conference* (pp. 12–24).

[2] Duce, F. et al. (2024). “You’ll Be a Nurse, My Son!” Automatically Assessing Gender Biases in Autoregressive Language Models in French and Italian. *Language Resources and Evaluation*.

[3] Gyax, P. M. et al. (2019). Exploring the Onset of a Male-Biased Interpretation of Masculine Generics Among French Speaking Kindergarten Children. *Frontiers in Psychology*, 10.

[4] Gastil, J. (1990). Generic Pronouns and Sexist Language. *Sex Roles*, 23(11–12).

[5] Misersky, J. et al. (2019). Grammatical Gender in German Influences How Role-Nouns Are Interpreted. *Discourse Processes*, 56(8).

[6] Doyen, E., & Todirascu, A. (2025). Man Made Language Models?. arXiv.

[7] Schmidt, J., & Hess, C. (2026). Tackling the Generic Masculine. *International Conference on Gender Research*, 9(1).

[8] Doyen, E., & Todirascu, A. (2025). GeNRe. In *Findings of the Association for Computational Linguistics: ACL 2025* (pp. 7889–7909). Vienna, Austria: Association for Computational Linguistics.

[9] Veloso, L. et al. (2023). A Rewriting Approach for Gender Inclusivity in Portuguese. In *Findings of the Association for Computational Linguistics: EMNLP 2023* (pp. 8747–8759). Singapore: Association for Computational Linguistics.